

Depth-Aware Dynamic Projection Mapping using High-speed RGB and IR Projectors

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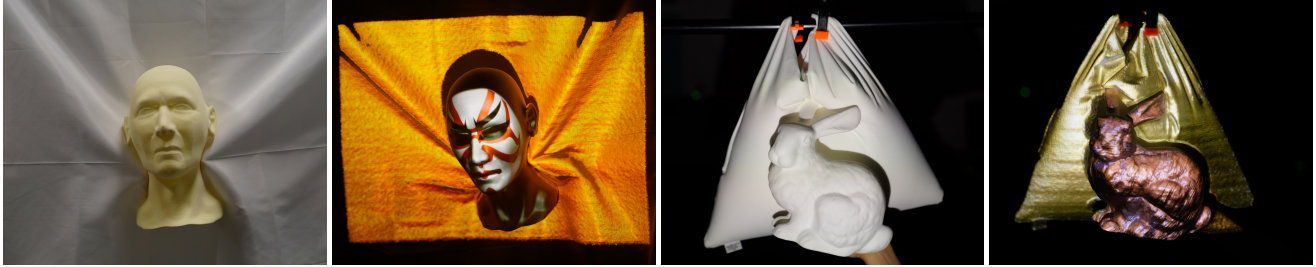


Figure 1: Our system can provide the depth-aware mapping of an entire scene using a projection that mixes the tracking-based UV-texturing image and the modelless shading image.

ABSTRACT

In this paper, we propose a strong system for dynamic projection mapping (DPM). It consists of a high-speed 500-fps camera, high-speed 24-bit 947-fps RGB projector, and newly developed 8-bit 2,880-fps infrared (IR) projector. This configuration allows us to capture the depth map and project the depth-aware image at a high frame rate. We also realize 0.4-ms markerless 3D pose tracking using the depth map by leveraging small inter-frame motions under high-frame-rate capture. We exploit these captured data and apply the depth-aware DPM to an entire scene with low latency, in a setting in which tracking-based and modelless mapping are combined and performed simultaneously.

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1 INTRODUCTION

With advancements in augmented reality technology, a diverse set of demands has emerged in terms of its capabilities. One of the core goals for augmented reality is to provide an immersive experience without the need to wear special devices. Dynamic projection mapping (DPM) is a key base technology that can be used to meet such demands [Grundhöfer and Iwai 2018]. DPM inherently requires high speed in capturing, rendering, and projection (at the level of 500 - 1,000 fps) to avoid perceived misalignments due to the large latency. Although several conventional works have recently made it possible to meet such requirements, they introduce some limitations in exchange for the high-speed performance. For example, the target must have a marker on the surface to improve the tracking speed [Narita et al. 2017]. In a markerless approach, the target is limited to a plane [Kagami and Hashimoto 2019] or face [Bermano et al. 2017]. In markerless and modelless cases, UV texturing cannot be fully implemented [Miyashita et al. 2018].

The DPM ideally needs to cover an entire scene by projection. A scene consists of a UV-textured surface and another surface, analogous to the process of graphics rendering. A UV-textured surface requires tracking, and it has practical benefits if it can be achieved

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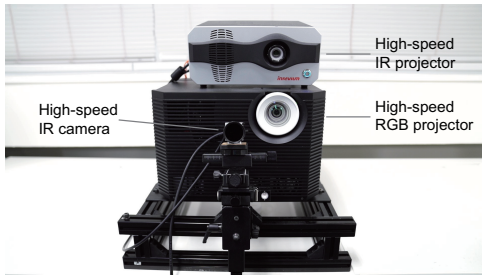


Figure 2: System configuration.

using an easy-to-prepare method. Under these requirements, DPM is desirable to implement both tracking-based markerless mapping and modelless mapping. A simple but effective solution is to obtain the depth map of the scene with high accuracy at a high speed, apply millisecond-order tracking, separate the scene into a UV-textured area and other areas, render the augmented appearance for each area, and project the image by mixing them with low latency. We call this approach the depth-aware DPM. However, it is difficult for a system with an off-the-shelf consumer sensor to perform such a DPM because it cannot realize depth capturing and 3D tracking at a speed that meets the requirements of the DPM.

We report a new DPM system that combines a 947-fps 24-bit high-speed RGB projector [Watanabe and Ishikawa 2019], 500-fps high-speed infrared (IR) camera, and newly developed 2,880-fps 8-bit IR high-speed projector. In addition, we have developed a high-speed model-based rigid-body tracking method that uses a depth map captured by a high-speed structured-light method. Tracking can be performed at a 0.4-ms processing time by leveraging a small inter-frame motion. Our system demonstrates flexible, depth-aware DPM on the entire scene with a latency of approximately 8 ms.

2 METHOD

As shown in Figure 2, our system consists of RGB and IR high-speed projectors and a high-speed IR camera with a 1/2.9" sensor. Our system captures a depth map using the IR ProCam pair and projects a depth-aware DPM image using the RGB projector. Specifically, the 8-bit IR high-speed projector is a unique and original device because the power of IR light sources are typically low to the extent that it is difficult to develop even a normal, low-speed projector. We overcame this limitation by customizing the projector with the latest strong light source, which is not specialized for projectors.

Depth sensing is based on phase shifting [Srinivasan et al. 1984] and works by projecting and capturing sinusoidal and gray-code patterns alternately in sequence at 500 fps in the IR range. A depth map computed on the GPU for every frame by using the latest six frames is fed into the tracking step, which is also computed on the GPU. In this study, the tracking method assumes the target to have a rigid body. Our method alters the localization step in *KinectFusion* [Newcombe et al. 2011]. As with the referenced method, the computation is parallelized based on the projective data association algorithm. Alternatively, the sensing frame rate in our system is much higher than that in the conventional system. Therefore, the inter-frame motion can be assumed to be small to an extent that it is within several pixels. We leverage this advantage to simplify the method. Specifically, we apply each data association and pose

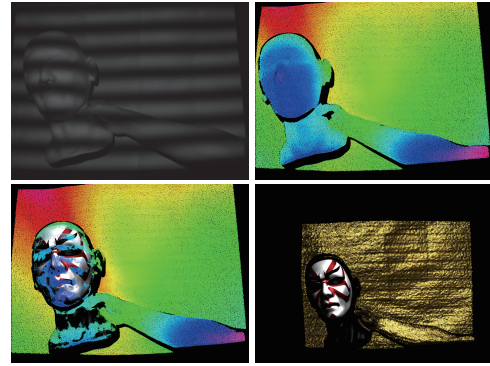


Figure 3: Captured image (left, above). Acquired depth map (right, above). Tracking result (left, bottom). Projected image in the projector coordinate system (right, bottom).

minimization only once. We also do not introduce a coarse-to-fine framework. As a result, the computation time is less than 0.4 ms on a computer with an Intel Xeon Gold 6136 and NVIDIA GeForce RTX 3090 while maintaining robust tracking.

For the scene to be rendered, we arrange a 3D model with UV-texture information based on the tracked pose results. We also arrange the vertex and normal data from the captured depth for the remaining surface. By exploiting this information, graphics are rendered flexibly.

3 RESULTS

Figure 3 shows the captured image, the depth map, tracking result, and projected image. Figure 1 shows the mapping results. Our system can perform projection mapping on the entire scene, which consists of the UV-texturing surface and the other surface. The total system latency from the motion to projection is approximately 8 ms. The current bottleneck in the proposed system is the camera frame rate and the depth noise, but we expect that this can easily be overcome by introducing a higher-frame-rate camera with a larger sensor size.

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